Computer Forensics - Quick Exam Notes

Quick Reference | All Topics Covered

*Brought to you by **ZenYukti***

1. Introduction & Digital Forensics Science

What is Computer/Digital Forensics?

- Application of science to identify, collect, examine & analyze digital data
- Preserves evidence integrity for legal proceedings
- Also called: Cyber Forensics / Digital Forensics

Digital Forensics Science

- Systematic approach to recovering digital evidence
- Must be: Authentic, Accurate, Complete, Convincing, Admissible
- Uses scientific methods — reproducible, documented, verifiable

Why Computer Forensics is Needed

- Cybercrime is increasing (hacking, fraud, data theft, IP theft)
- Courts require digital evidence to be properly collected & presented
- Helps in: criminal investigations, civil litigation, corporate investigations
- Data recovery after accidents/disasters

2. Cyber Forensics & Digital Evidence

Digital Evidence

- Any data stored/transmitted digitally that can be used in court
- Types: files, emails, logs, metadata, browser history, database records
- Must satisfy: Integrity (unchanged), Authenticity (from where claimed), Reliability

Key Properties of Evidence

- Admissible — legally acceptable in court
- Authentic — provably from claimed source
- Complete — full picture, not cherry-picked
- Reliable — consistent across examinations
- Believable — understandable to jury/judge

Forensic Analysis of E-Mail

- Header analysis → reveals sender IP, mail servers, timestamps
- Email headers contain: From, To, Date, Message-ID, Received (routing path)
- Tools: Email tracer, eMailTrackerPro, Forensic Email Collector
- Spoofed emails: check X-Originating-IP in headers
- Deleted emails can be recovered from server logs or PST/OST files

3. Digital Forensics Life Cycle

- Step 1 — Identification: Recognize potential sources of evidence (devices, media)
- Step 2 — Preservation: Isolate & protect data (write-blockers, evidence bags)
- Step 3 — Collection: Acquire forensic copies (bit-by-bit/disk images)
- Step 4 — Examination: Filter & extract relevant data
- Step 5 — Analysis: Interpret data, reconstruct events, find patterns
- Step 6 — Presentation: Report findings in court-ready, understandable format
- Step 7 — Decision/Reporting: Document all steps, tools used, findings

Think of it like: IPCEAS → Identification, Preservation, Collection, Examination, Analysis, Presentation

4. Chain of Custody Concept

- Documented history of who handled evidence, when, where, and why
- Critical for evidence admissibility in court
- Any break = evidence may be declared inadmissible
- Must record: Collection time/date, who collected, storage location, transfers
- Tamper-evident seals used on physical media
- Hashing (MD5/SHA-256) used to prove digital evidence not altered
- Form documents every access to evidence

5. Network Forensics

Definition

- Monitoring, capturing, storing & analyzing network traffic for evidence
- Two approaches:
  - Catch-it-as-you-can: Capture all packets, analyze later (storage-heavy)
  - Stop-look-listen: Analyze on the fly, only store suspicious data

Key Concepts

- Log files: Firewall logs, IDS logs, web server logs, DHCP logs
- Packet captures: Wireshark, tcpdump
- IDS/IPS logs — intrusion detection evidence
- MAC address & IP address tracing for attribution
- Network forensics helps detect: DoS, unauthorized access, data exfiltration

Tools

- Wireshark — packet analysis
- NetWitness — network forensics platform
- Xplico — network traffic decoder

6. Approaching a Computer Forensics Investigation

- Step 1 — Secure the scene: Prevent tampering, isolate devices
- Step 2 — Document everything: Photos, sketches, notes before touching anything
- Step 3 — Identify evidence: List all devices (PCs, phones, USBs, routers)
- Step 4 — Acquire image: Use write-blockers; create forensic bit-copy of drives
- Step 5 — Hash verification: MD5/SHA hash before and after — must match

- Step 6 — Analyze copy: NEVER work on original — always use forensic copy
- Step 7 — Document findings: Timeline of events, file access times, logs
- Step 8 — Report: Expert-witness ready report with methodology & findings

Important Principles

- Never modify original evidence
- Work on verified forensic copies (disk images)
- All actions must be repeatable and documented
- Use court-accepted tools (EnCase, FTK, Autopsy)

7. Forensics & Social Networking Sites

Security / Privacy Threats

- Identity theft — fake profiles, phishing via social platforms
- Cyberbullying & cyberstalking — harassment via DMs/posts
- Social engineering — attackers gather OSINT from public profiles
- Data leakage — employees unknowingly share confidential info
- Malware spread — malicious links shared via social media
- Account hijacking — password attacks, session hijacking
- Privacy violation — location tagging, check-ins exposing real-time location

Forensics on Social Media

- Evidence can be: posts, DMs, check-ins, friend lists, activity logs
- Request data from platform via legal process (subpoena/warrant)
- Screenshot + metadata capture for preservation
- Social media platforms may delete content — act fast
- IP logs from platform can identify real user location
- Tools: Belkasoft Evidence Center, Social Media Workbench

8. Challenges in Computer Forensics

- Encryption — BitLocker, AES encryption makes data inaccessible without keys
- Anti-forensics — wiping tools (Eraser), steganography, log tampering
- Volume of data — terabytes of data; hard to analyze everything
- Cloud computing — data across jurisdictions, provider cooperation needed
- Volatile data — RAM data lost on power-off; must capture first
- Cross-jurisdiction — cybercrime crosses national borders, legal complexity
- Device diversity — smartphones, IoT, wearables — each needs different tools
- Rapidly changing tech — forensic tools lag behind new OS/file system updates
- Legal/privacy issues — balance between investigation & privacy rights
- Admissibility — improper procedure = evidence thrown out in court

QUICK RECALL: Life Cycle = ID → Preserve → Collect → Examine → Analyze → Present | Chain of Custody = who touched it + when | Hash = proof of integrity | Never touch originals!

END OF NOTES

ZenYukti

Quick Exam Notes

zenyukti.in

Other Unit Notes

All subject notes on GitHub — free to access & star

github.com/ZenYukti/MyNotes

Join ZenYukti Discord

Free workspace — internship & hackathon updates, tech discussions

go.zenyukti.in/discord

FIND US ON

Made with care by **ZenYukti** · Share freely, learn together

Quick Exam Notes | Exam Ready | Free Resource